



Selectivity setup in the North Sea sole SS3 model

ICES WGNSSK 2026

21-30 April 2026 | Iago Mosqueira (WMR), Alessandro Orio (SLU) (Wageningen Marine Research)

Problems with current setup

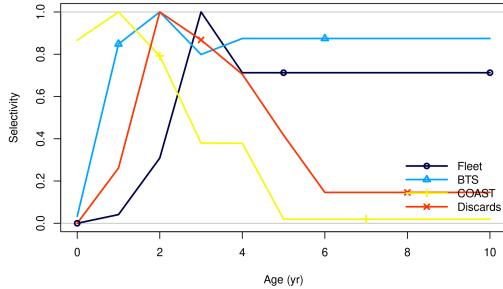
- Current flexibility uses random walks for all fleets, landings changing on time
- Set to a constant value from a given age
- Landings selectivity freely estimated in time
- Final years commonly showing drops in selectivity at age 3-4
- Number of parameters too high

Changes to selectivity setup

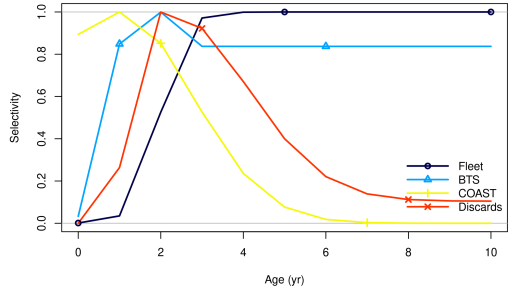
- Random walk per year, fixed from age 5
- Moved to functional forms for all fleets
- Fix a few poorly-estimated parameters
- **Landings**, Logistic
- **BTS** Double normal
 - p4 (descending width), p6 (final), avoid drop at older ages
 - Allowed to vary on pulse years, 2009-2021
- **COAST** Double normal
 - p1 (peak) fixed at ~ 0
- **Discards** Double normal

Estimated selectivities

- WGNSSK

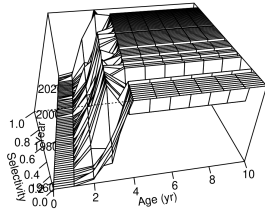


- SELEX

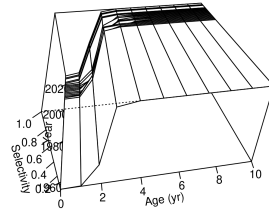


Landings selectivity

- WGNSSK

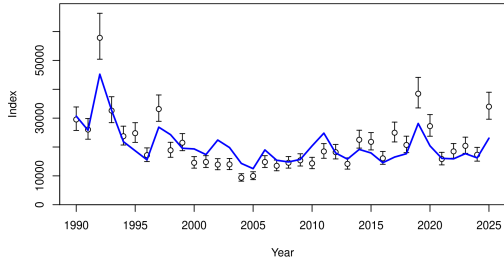


- SELEX

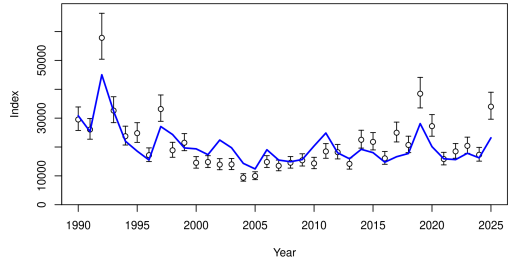


Fit to BTS

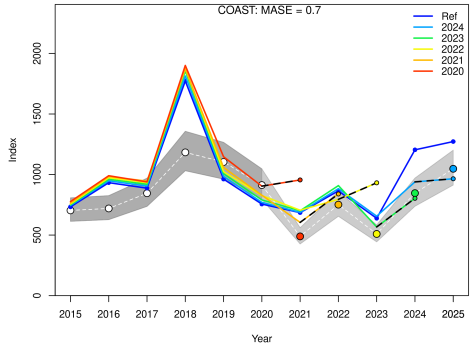
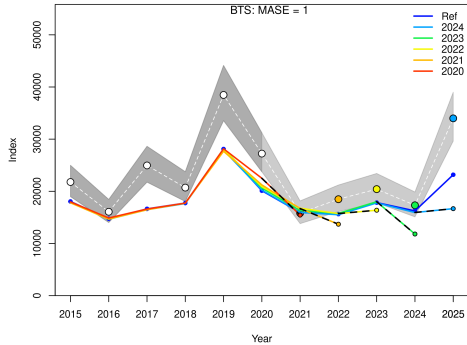
- WGNSSK



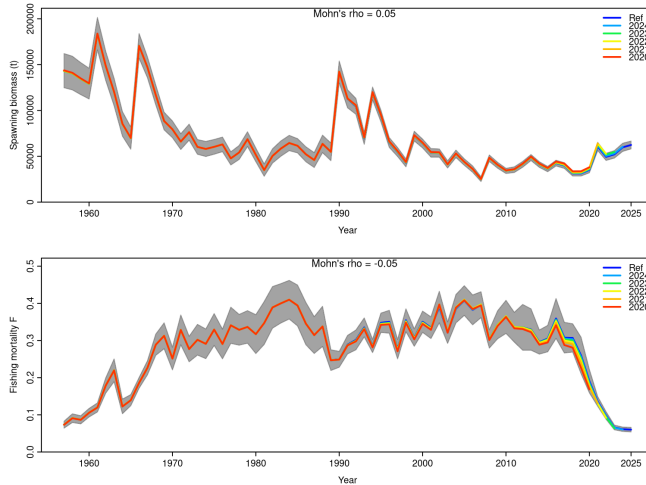
- SELEX



Prediction skill

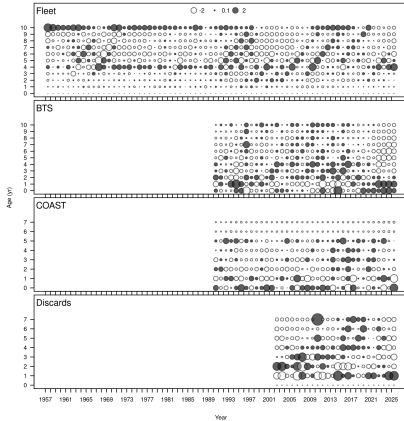


Retrospective patterns

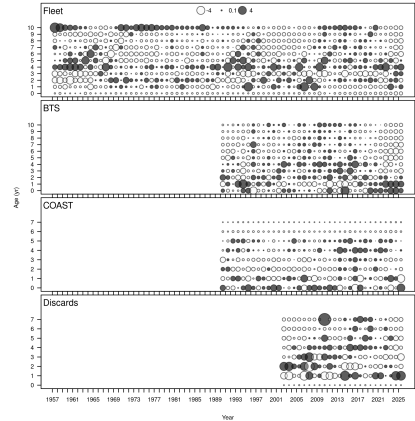


Residuals

- WGNSSK



- SELEX



Setup

#	L0	HI	INIT	PRIOR	PR_SD	PR_type	PHASE	env-var	use_dev	dev_mnyr	dev_mxyr	dev_PH	Block	BFxn	parm_name
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---
# 1 Fleet AgeSelex															
0	20	2.23076	10	4	0.5	5	0	23	2009	2021	5	0	0		Age_inflection_Fleet(1)
-4	30	0.710311	1	4	0.5	6	0	23	2009	2021	6	0	0		Age_95%width_Fleet(1)
# 2 BTS AgeSelex															
0	9.9	1.2036	2	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_peak_BTS(2)
-5	3	-3.79741	-5	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_top_logit_BTS(
-4	12	-1.44672	8.4	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_ascend_se_BTS(
-10	6	-3.47201	3	1	0.05	-4	0	0	0	0	0.5	0	0		Age_DblN_descend_se_BTS
-15	5	-3.38987	0	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_start_logit_BT
-5	15	1.63717	0.9	1	0.05	-4	0	0	0	0	0.5	0	0		Age_DblN_end_logit_BTS(
# 3 COAST AgeSelex															
0	9.9	4.32E-05	2	1	0.05	-4	0	0	0	0	0.5	0	0		Age_DblN_peak_COAST(3)
-15	3	-12.5387	-5	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_top_logit_COAS
-30	12	-20.9735	5.9	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_ascend_se_COAS
-2	6	1.82926	0.3	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_descend_se_COA
-15	5	1.31219	0.8	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_start_logit_CO
-15	5	-11.8416	0	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_end_logit_COAS
# 4 Discards AgeSelex															
0	9.9	1.19921	2	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_peak_Discards(
-15	3	-10.277	-5	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_top_logit_Disc
-4	12	-3.59095	0.3	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_ascend_se_Disc
-2	6	1.956	1	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_descend_se_Dis
-15	5	-14.6525	0	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_start_logit_Di
-5	5	-2.14541	0.2	1	0.05	4	0	0	0	0	0.5	0	0		Age_DblN_end_logit_Disc

Estimated parameters

- WGNSSK

	Type	Count
1	Total	330
7	Stock-recruit	1
9	Rec. dev. time series	69
10	Rec. dev. initial age	8
18	Age selectivity	18
19	Age selectivity time-variation	207
20	F parameters time series	24
21	F parameters initial equilibrium	1
22	Dirichlet-Multinomial parameters	2

- SELEX

	Type	Count
1	Total	148
7	Stock-recruit	1
9	Rec. dev. time series	69
10	Rec. dev. initial age	8
18	Age selectivity	17
19	Age selectivity time-variation	26
20	F parameters time series	24
21	F parameters initial equilibrium	1
22	Dirichlet-Multinomial parameters	2

Summary

- Proposed less flexible selectivity setup
- Fewer parameters, more stable estimates
- Fit not dramatically altered
- Are recent fishery changes in need of another block?
- What is happening with BTS?

https://github.com/iagomosqueira/2026_sol.27.4_review/selectivity

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